

Proceedings of the National Academy of Sciences of the United States of America (PNAS)

Editorial Decision Letter

Manuscript ID: PNAS-2026-08473

Manuscript Title: Modulation of oxidative stress and inflammatory signaling in cellular models of age-associated dysfunction

Article Type: Research Article

Decision: Major Revision

Dear Dr. Martínez,

Thank you for submitting your manuscript to PNAS. Your study addresses an important biological question concerning the relationship between oxidative stress, chronic inflammatory signaling, and age-associated cellular dysfunction. The reviewers found the topic timely and potentially relevant to a broad readership interested in mechanisms of aging, inflammation, and translational biomedicine.

However, after consultation with the reviewers, we cannot consider the manuscript further in its present form. We would be willing to evaluate a substantially revised version, provided that the authors address the major conceptual, methodological, and interpretative concerns outlined below. Please note that a revised manuscript would not be guaranteed acceptance and may be returned to the original reviewers.

Summary of the evaluation

The reviewers agreed that the manuscript is clearly written and that the experimental question is of interest. The preliminary evidence suggesting that compound X reduces reactive oxygen species and modulates inflammatory markers is potentially valuable. Nevertheless, the reviewers raised concerns about the strength of the mechanistic claims, the limited description of experimental replication, the incomplete statistical reporting, and the current framing of the work for a broad PNAS readership.

Major comments

1. Conceptual framing and significance

The manuscript would benefit from a clearer articulation of the central biological advance. At present, the study is framed as having broad relevance to aging and age-associated disease, but the evidence remains largely confined to an in vitro cellular model. The authors should clarify what is genuinely new relative to previous work on antioxidant modulation of inflammatory pathways and explain why the findings should be of interest beyond the immediate subfield.

2. Mechanistic interpretation

Several statements in the Abstract, Discussion, and final paragraph imply that compound X may have therapeutic potential in age-associated disease. The reviewers considered these claims premature. The data support a more cautious conclusion: compound X appears to reduce oxidative stress and selected inflammatory readouts under the tested conditions. The authors should avoid therapeutic language unless additional validation is provided.

3. Experimental design and replication

The Methods section does not provide sufficient detail on biological replicates, technical replicates, passage number, cell source, donor variability, or exclusion criteria. The revised manuscript should explicitly report the number of

independent experiments, the number of wells or samples per condition, and how variability between experiments was handled.

4. Controls and dose-response analysis

The reviewers requested clearer justification for the three concentrations of compound X. A vehicle control should be explicitly described. If available, the authors should include a positive control for oxidative stress reduction or anti-inflammatory activity. The dose-response relationship should be presented more clearly, including whether any cytotoxicity was observed at higher concentrations.

5. Statistical reporting

The manuscript reports statistical significance but does not consistently provide exact p-values, effect sizes, confidence intervals, or the statistical test used for each comparison. The authors should revise the statistical reporting throughout the Results and figure legends. If multiple comparisons were performed, the correction method should be specified.

6. Figures and legends

The figures are generally understandable, but the legends do not consistently report n, the meaning of error bars, statistical tests, or whether data correspond to technical or biological replicates. Some panels combine descriptive and interpretative statements. The authors should revise figure legends so that each figure can be interpreted independently from the main text.

7. Limitations

The Discussion does not sufficiently acknowledge limitations. In particular, the authors should discuss the limitations of the in vitro model, the absence of independent validation, the lack of direct pathway inhibition or rescue experiments, and the limited basis for translational claims.

8. Data availability and transparency

The Data Availability statement is too generic. The authors should specify where raw data, processed data, and analysis scripts will be made available. If code was not used beyond standard statistical software, this should be stated explicitly.

Minor comments

1. The title may overstate the breadth of the findings. Please consider a more precise title.
2. The Significance Statement should be rewritten for a broad scientific audience and should avoid repetition of the Abstract.
3. Several abbreviations are introduced inconsistently, including ROS and TNF-alpha.
4. The Introduction contains some general statements about aging that would benefit from more precise referencing.
5. Please ensure that all ethical and funding statements are complete.

Instructions for revision

Please submit a revised manuscript together with a detailed point-by-point response to each reviewer and editorial comment. In your response, indicate exactly where changes have been made in the manuscript. Where you disagree with a comment, provide a clear scientific justification. Please also include a marked version of the manuscript showing all changes.

We encourage the authors to focus the revision on strengthening the mechanistic interpretation, improving methodological transparency, moderating claims, and clarifying the broader significance of the work.

Sincerely,

Dr. Elizabeth R. Coleman
Associate Editor
PNAS